



Blood Test Abnormalities Screening

Machine Learning Modeling Using 2021-2023 NHANES Data
By: Dominic Sciarrino & David Graham



Table of contents

01

**Problem definition
and motivation**

02

**Data Collection
and Preprocessing**

03

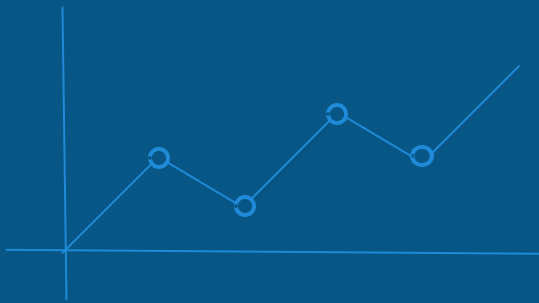
Modeling Approach

04

Model Evaluation

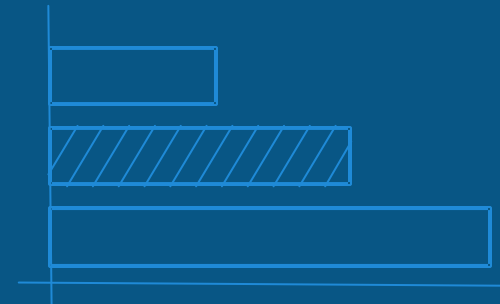
05

**Recommendations
and Prescriptive
Insights**



01

Problem definition and motivation



The Problem

- Many people unknowingly live with conditions detectable through routine bloodwork which can serve as early biomarkers for chronic diseases (e.g., elevated HbA1c as a precursor to type2 diabetes)
- Early detection enables intervention before long term damage occurs
- But bloodwork isn't always ordered until symptoms appear, leaving a screening gap

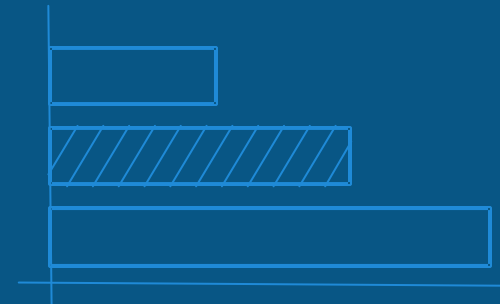
The Goal

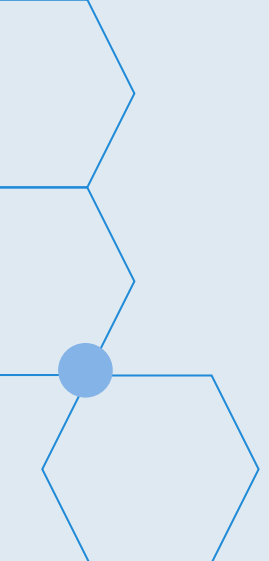
- Build a screening tool that flags individuals at elevated risk of abnormal blood results using examination data and lifestyle factors
- This is not a diagnostic tool. It is meant to facilitate conversations with qualified healthcare professionals to decide when targeted testing or intervention is needed



02

Data Collection and Preprocessing

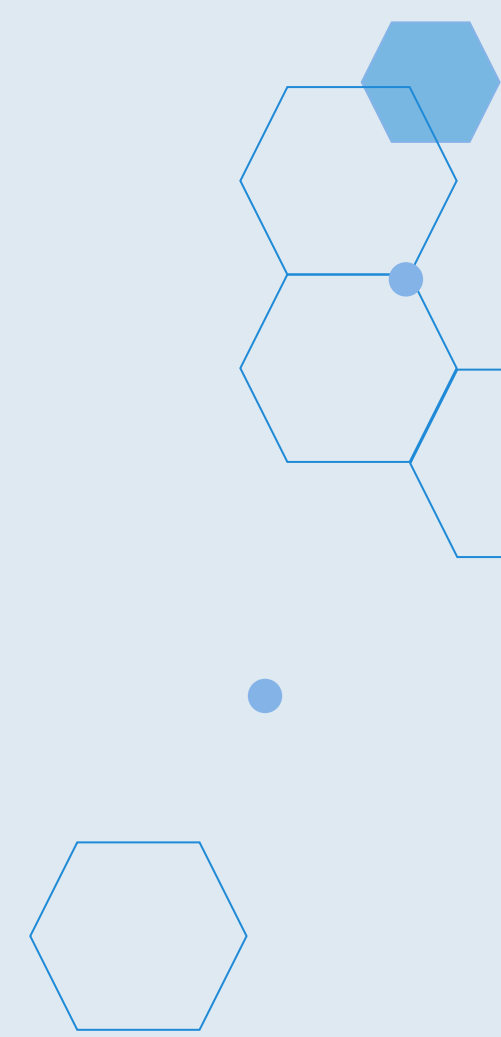




Data Source

2021-2023 National Health and Nutrition Examination Survey (NHANES)

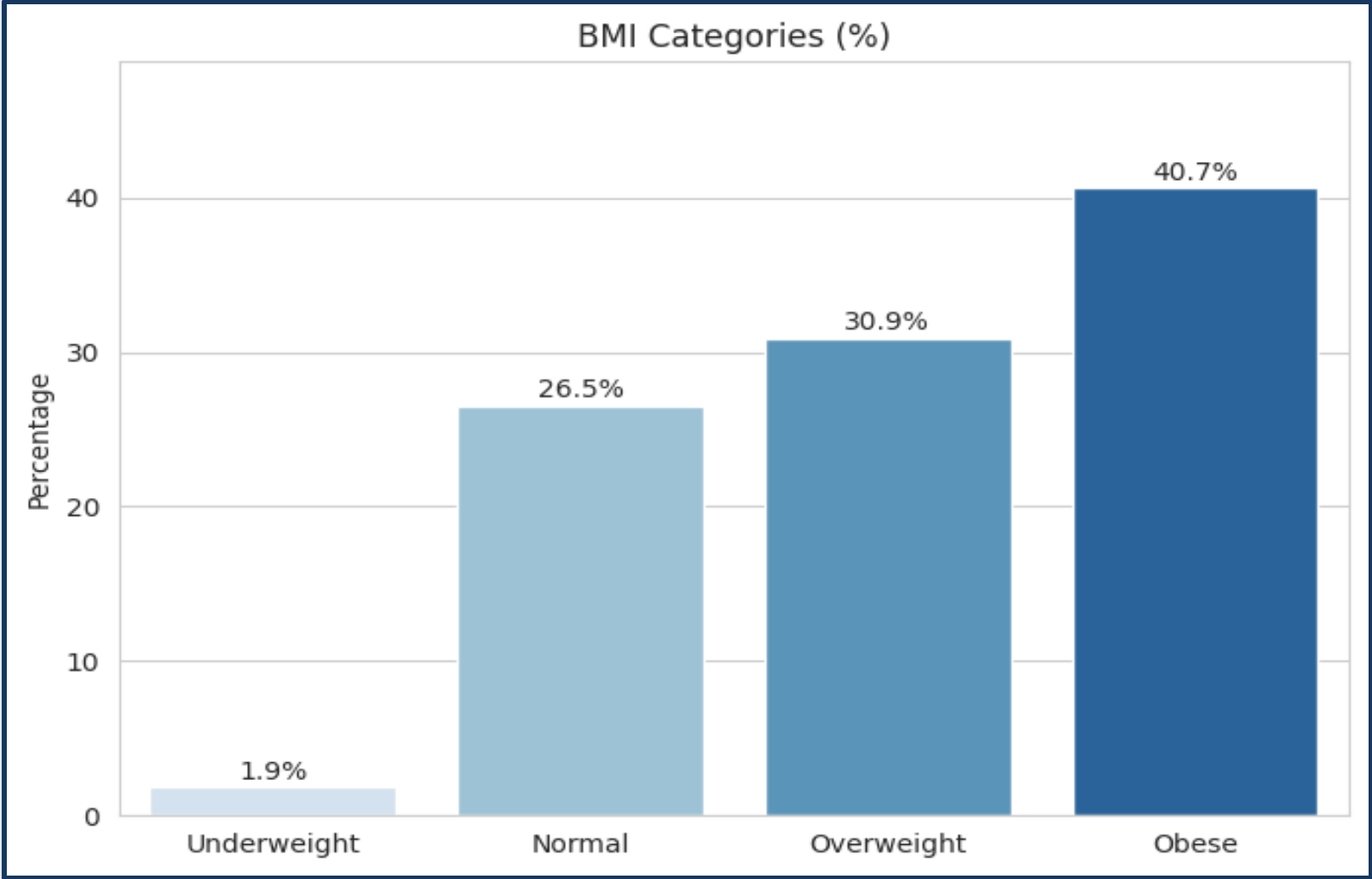
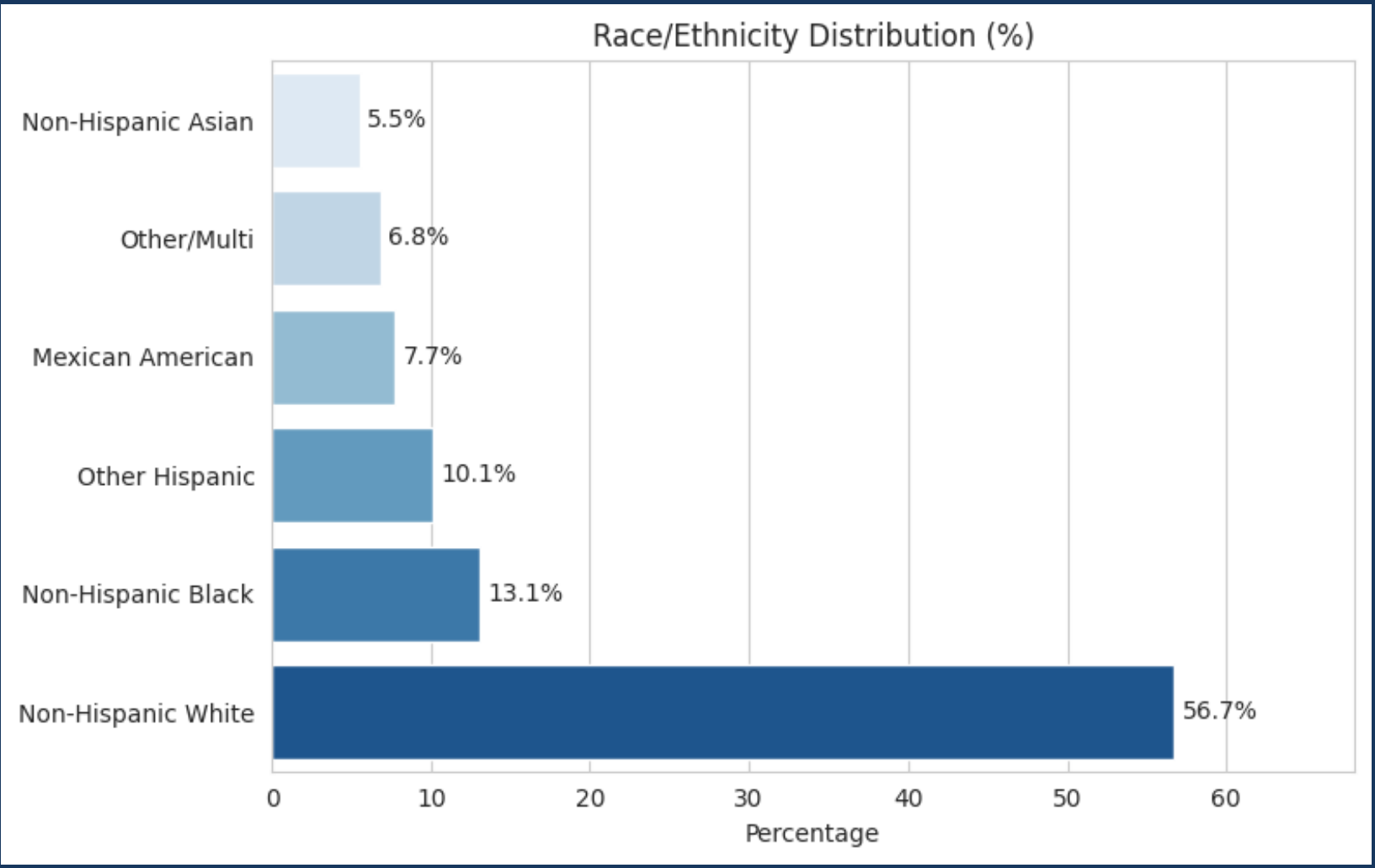
Data collected by the CDC's National Center for Health Statistics

- Features include participant answers about their demographics, diet, and other lifestyle factors, physical examination data (BMI, blood pressure, balance tests) and laboratory blood tests
 - Participants are selected using a complex, four-stage probability sampling design to ensures the selected individuals represent the entire U.S. population
 - About 12,000 participants over the 2-year period
- 

Data Overview

Final Dataset: 7,587 participants & 128 Features

Age (years)	
Mean $\pm$ SD	50.3 $\pm$ 17.6
Median (IQR)	53.0 (35.0–65.0)
Gender	
Female	54.9%
Male	45.1%



Challenges

Large Database

1

Built a user-friendly interface to explore the data

High dimensionality

2

1. Correlation Based Filtering
2. Boruta
3. Elastic Net Logistic Regression

Data Sparsity

3

Used KNN to impute NAs for model input, giving more accurate imputation than just median/mode.

Large Class Imbalance

4

- stratify=y preserves class balance in train and test sets
- BalancedRandomForestClassifier
- class_weight="balanced" for Logistic Regression

NHANES Explorer Tool

NHANES.explorer

 Search

 Library

 Labs

Variable Search

TF-IDF semantic search across 30 NHANES codebook variables

 Search

Showing 30 results for “sleep”

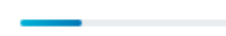
SLD012

CONTINUOUS

Questionnaire

Sleep hours - weekdays or workdays

Number of hours usually sleep on weekdays or workdays.

 score 0.4975 · SLQ_L



Details →

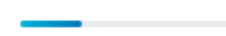
SLD013

CONTINUOUS

Questionnaire

Sleep hours - weekends

Number of hours usually sleep on weekends or non-workdays.

 score 0.4972 · SLQ_L



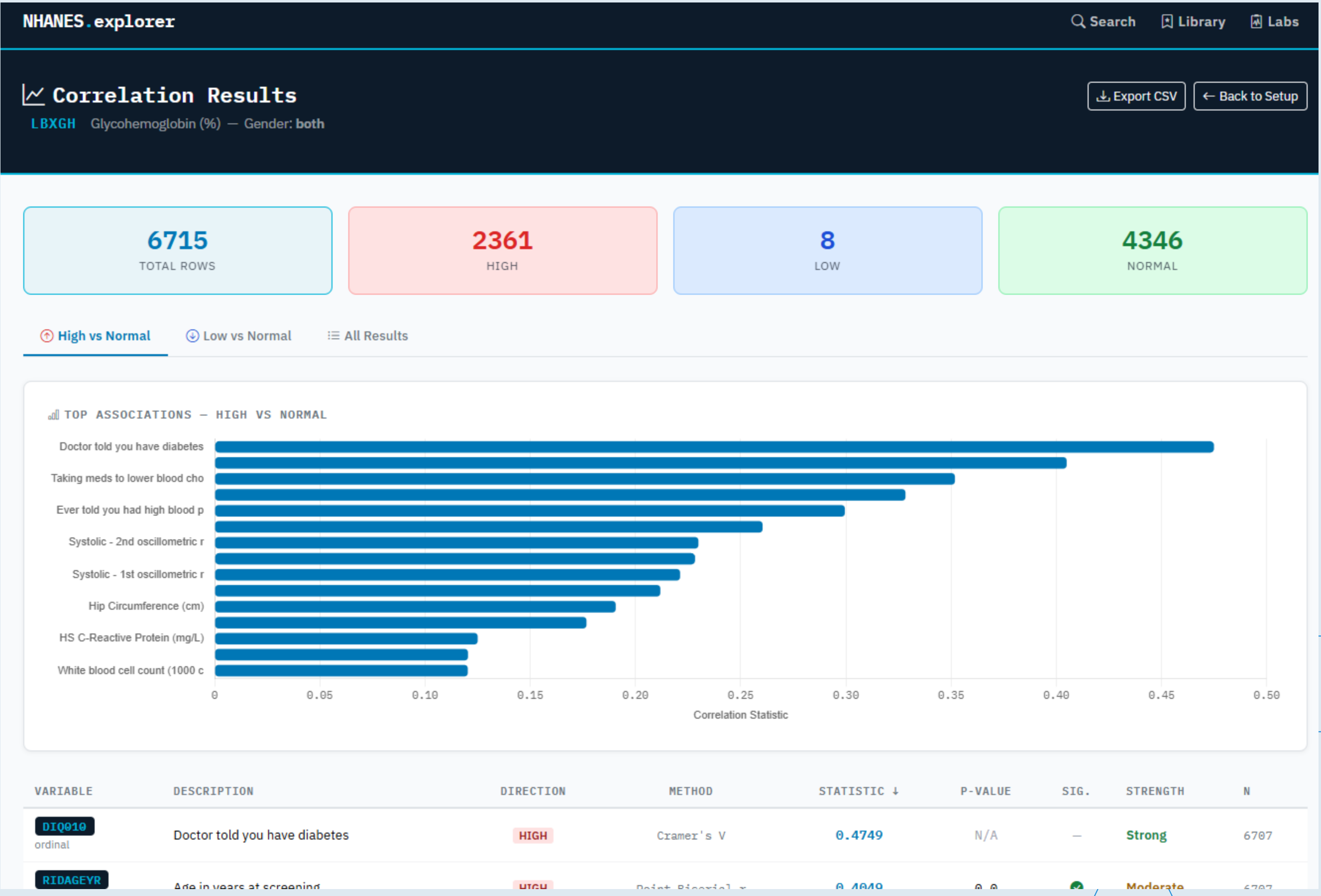
Details →

NHANES Explorer is a Django-based tool designed to make the NHANES dataset easier to navigate. Because the dataset uses coded variables, the tool includes a fuzzy search feature that lets users find relevant codes using plain-language terms. For example, searching “sleep” will return related variables even if that exact word isn’t used in the code names.

NHANES Explorer Tool

The Explorer also includes a correlation tool that helps identify features related to a chosen target variable.

It uses two methods: Cramer's V for categorical variables, and the point-biserial correlation for relationships between continuous and binary variables.





03

Modeling Approach

Model Pipeline

1

Data Preparation

Filter specific responses/features and engineer binary target (normal (0) vs abnormal (1))

2

Split Data

Use an 80/20 split to create train and test sets

3

Preprocessing

One hot encoding, scaling, KNN imputing

4

Feature Selection

Ran Boruta to reduce to about 10-20 features.

5

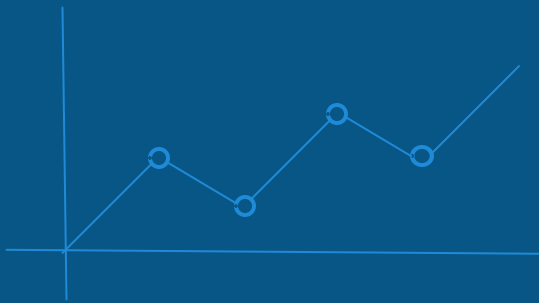
Fit Models

Random Forest and Elastic Net Logistic Regression

6

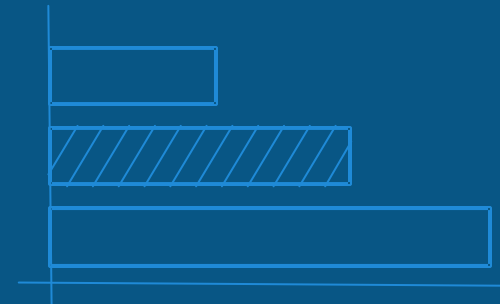
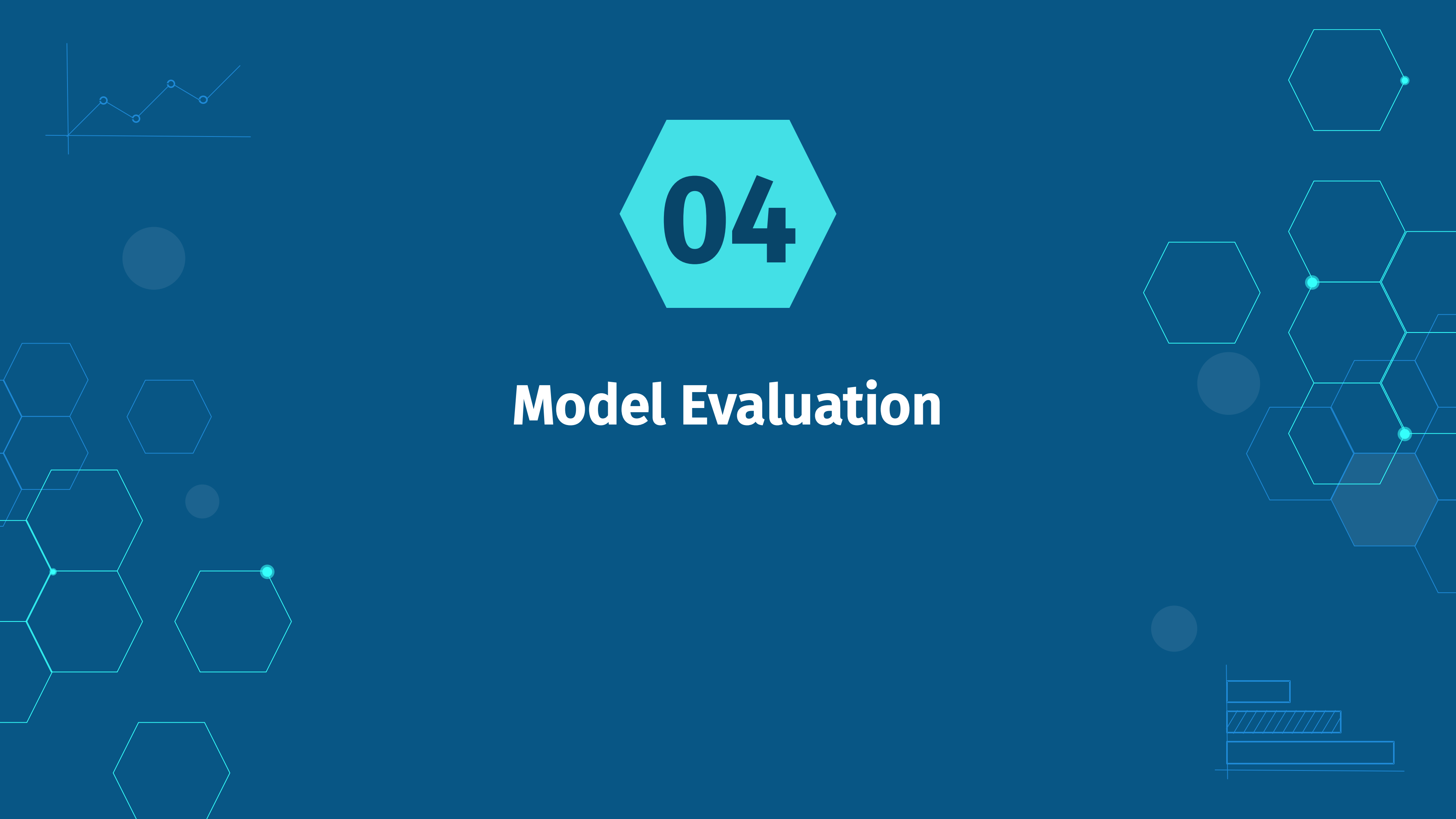
Model Evaluation

Evaluate based on AUC, positive recall and AUC gap metrics



04

Model Evaluation



Overall Evaluation

Evaluation metrics across 5 different models: Hba1c, Hscrp, Vitamin D, WBC, HDL

- Despite slightly lower recall Logistic Regression consistently produced smaller gaps between train and test AUC which implies it is better at generalizing and not overfitting. This is important for real world deployment
- Logistic regression also allows for easier interpretability of feature importance to translate into survey questions

1

Average Positive Recall

Random Forest: 0.69
Logistic Regression: 0.67

2

Average AUC

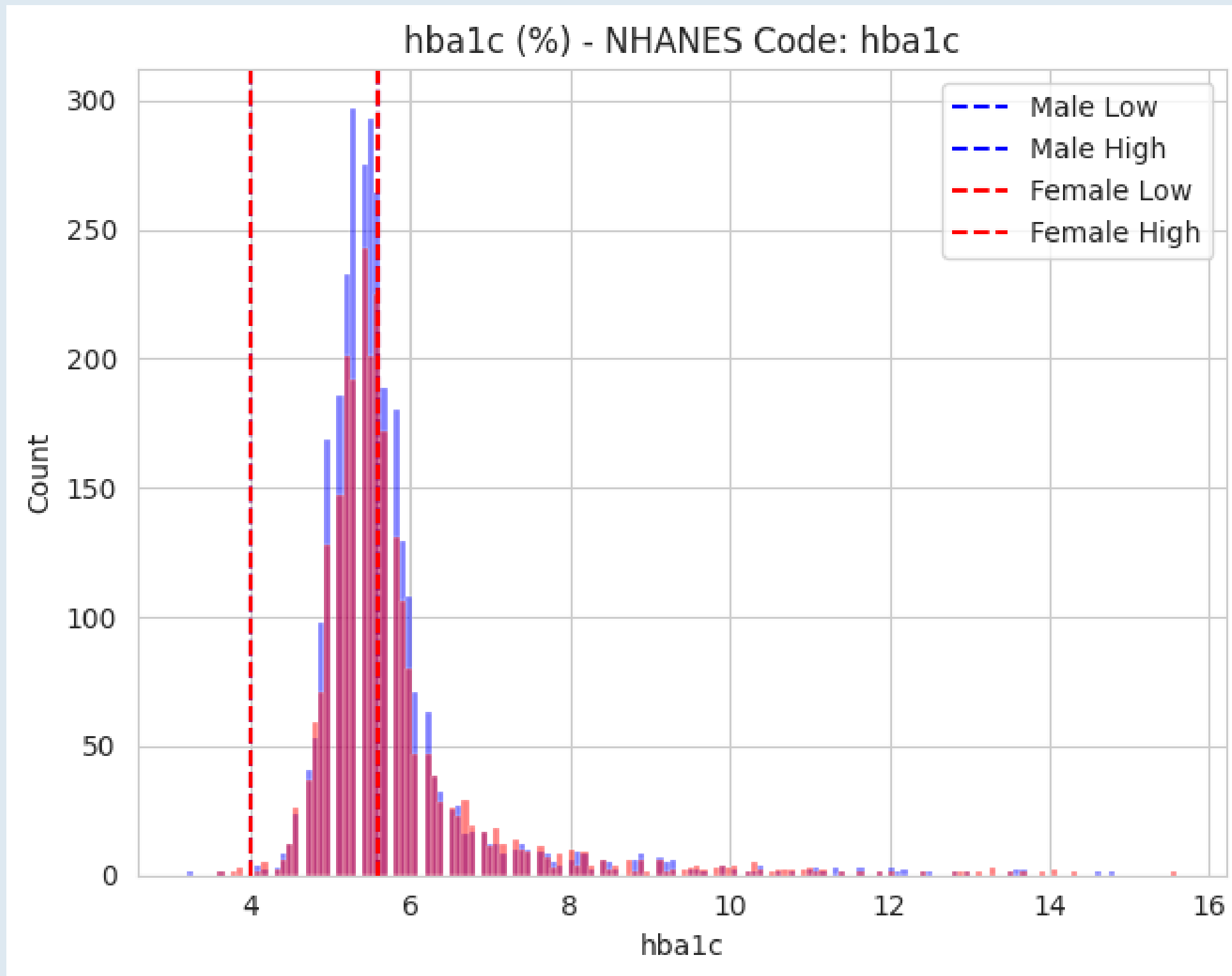
Random Forest: 0.75
Logistic Regression: 0.76

3

Average AUC Gap

Random Forest: 0.07
Logistic Regression: -0.0002

Model 1: High Hba1c

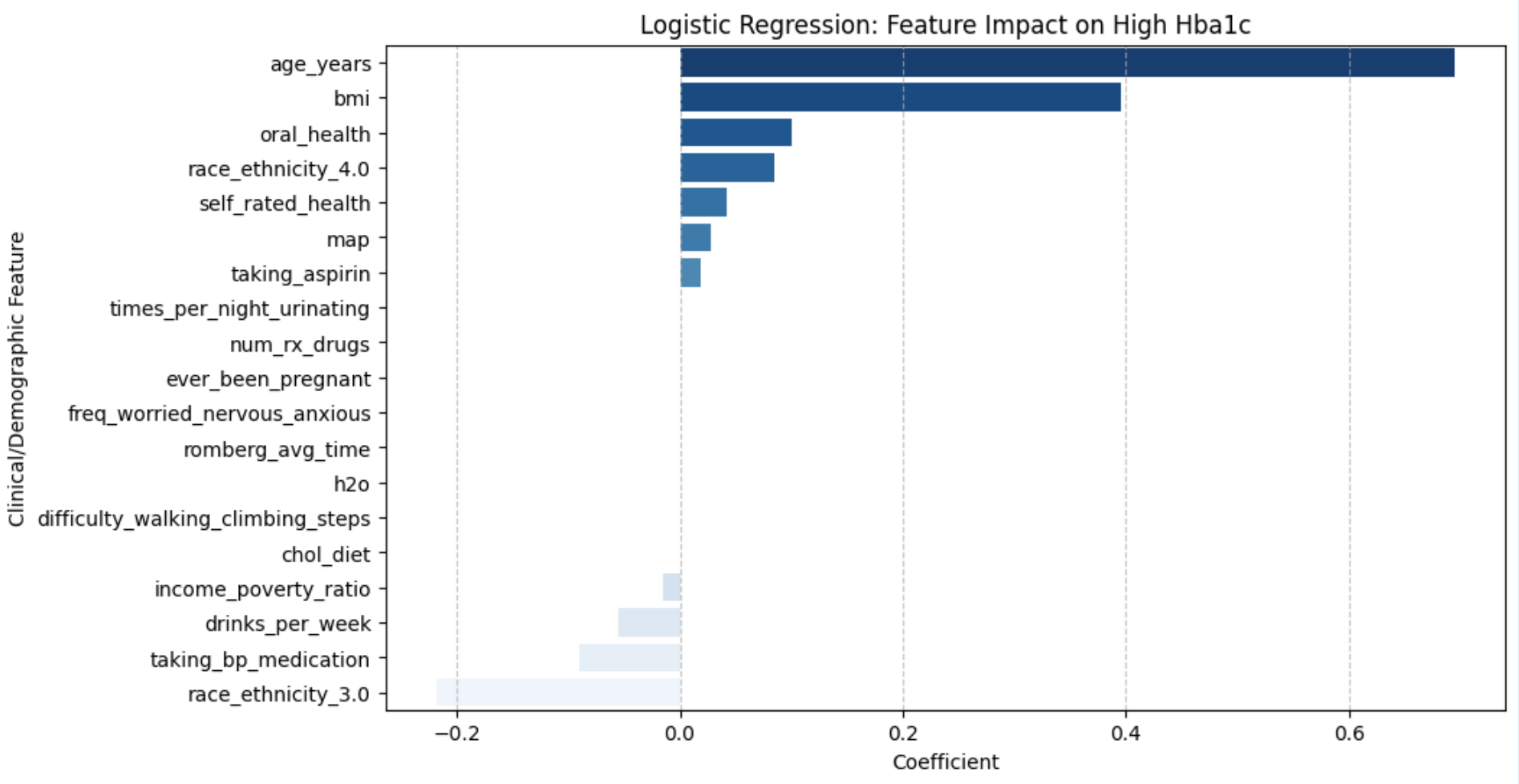


High Threshold: > 5.7

- Normal: 3793 (78.8%)
- High (> 5.7): 1020 (21.2%)
- Measures the average amount of sugar in blood over the past several months
- High a1c can be an indicator of prediabetes or Type 2 diabetes

<https://www.mayocliniclabs.com/test-catalog/overview/82080#clinical-and-interpretive>

Model 1: High Hba1c

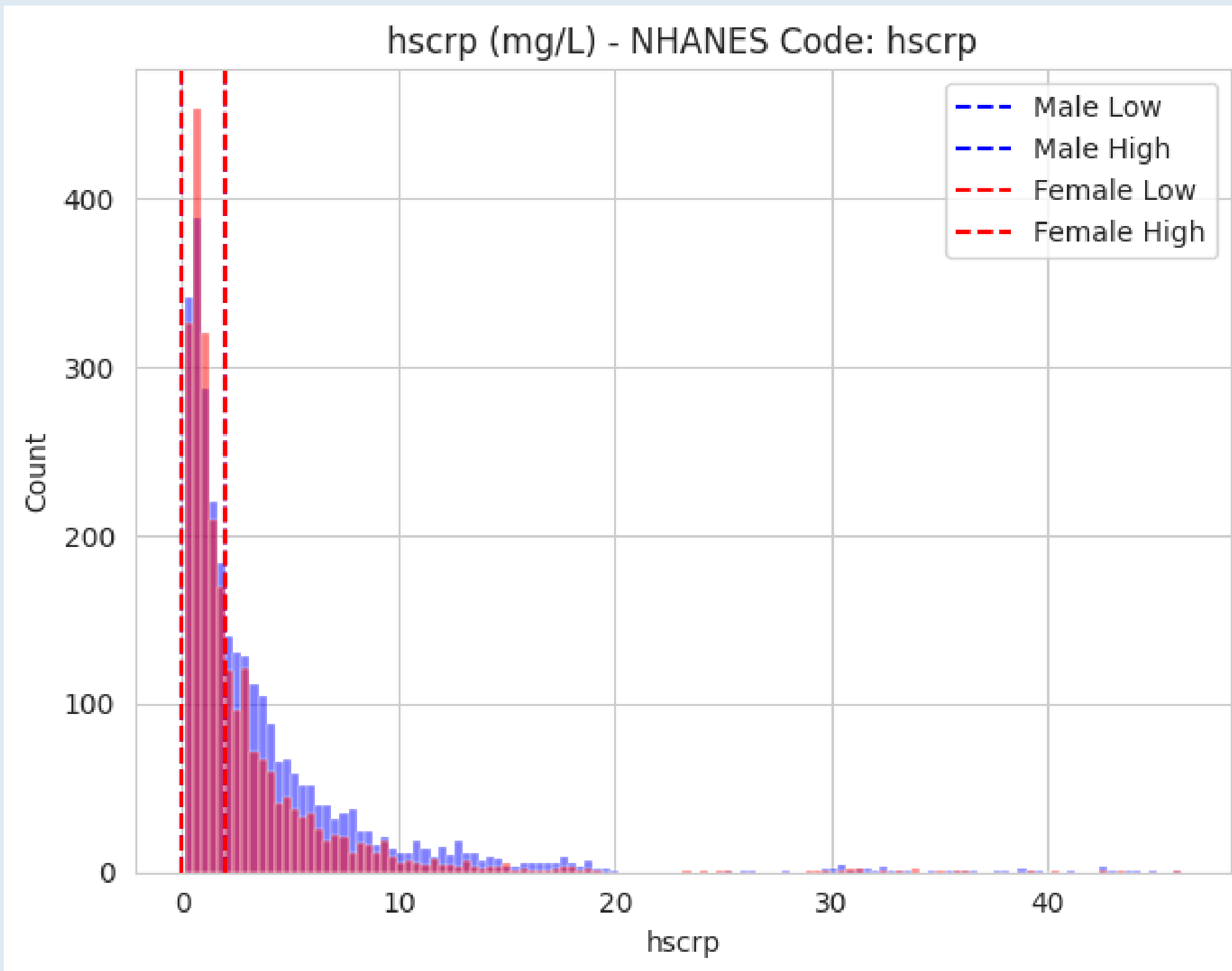


Top Predictors by odds ratio:

- Age: 2.00
- BMI: 1.49
- Oral Health : 1.11
- Non-Hispanic Black: 1.09
- Taking BP Medication (Yes=0): 0.91

Metric	Random Forest	Logistic Regression
Discrimination & Stability		
Test AUC	0.769	0.778
Train AUC	0.831	0.775
Train-Test gap	+0.062	-0.004
Detection of “High” (positive class)		
Recall	0.75	0.75
Precision	0.37	0.37
Detection of “Normal” (negative class)		
Recall	0.66	0.66
Precision	0.91	0.91

Model 2: High hsCRP

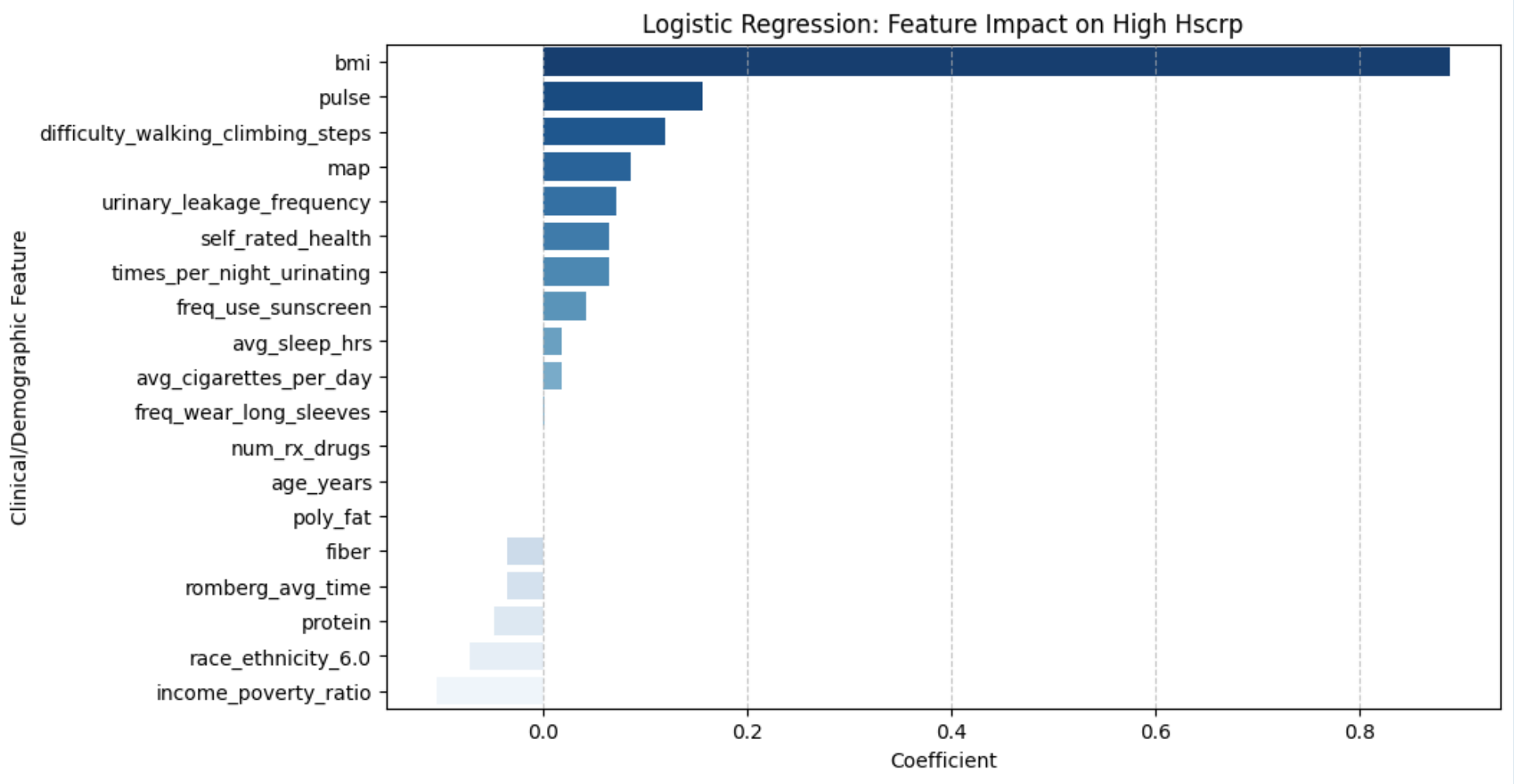


High Threshold: > 2 mg/L

- Normal: 2908 (53.1%)
- High (≥ 2): 2569 (46.9%)
- Assesses risk of cardiovascular disease or event
- High levels are biomarker of cardiovascular disease or events risk

<https://www.mayocliniclabs.com/test-catalog/overview/606909#clinical-and-interpretive>

Model 2: High hsCRP

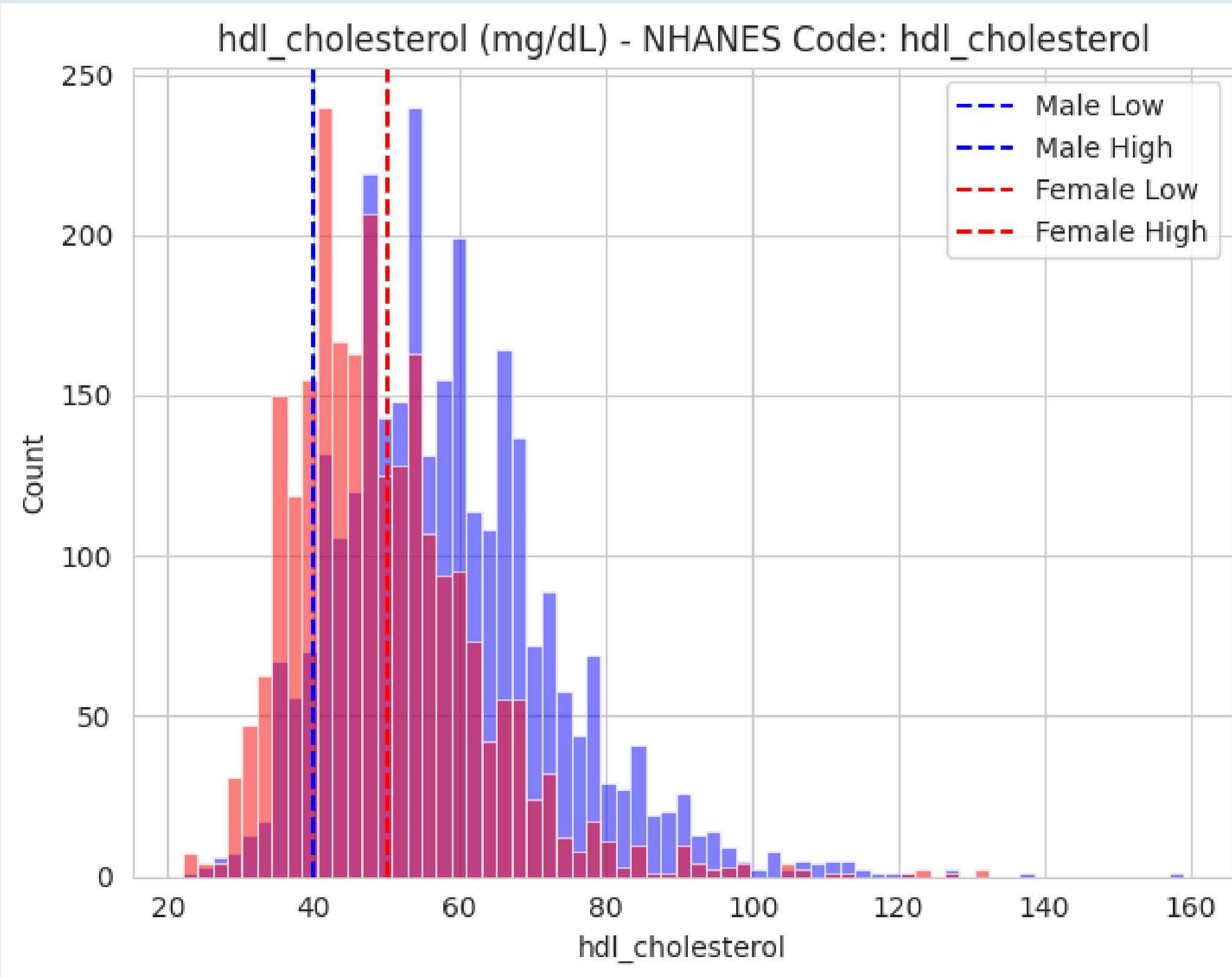


Top Predictors by odds ratio:

- BMI : 2.43
- Pulse: 1.17
- Difficulty walking/climbing steps: 1.13
- Blood Pressure: 1.09
- Income to Poverty Ratio: 0.90

Metric	Random Forest	Logistic Regression
Discrimination & Stability		
Test AUC	0.773	0.786
Train AUC	0.813	0.776
Train-Test gap	+0.040	-0.010
Detection of "High" (positive class)		
Recall	0.69	0.68
Precision	0.68	0.70
Detection of "Normal" (negative class)		
Recall	0.71	0.74
Precision	0.72	0.73

Model 3: Low HDL

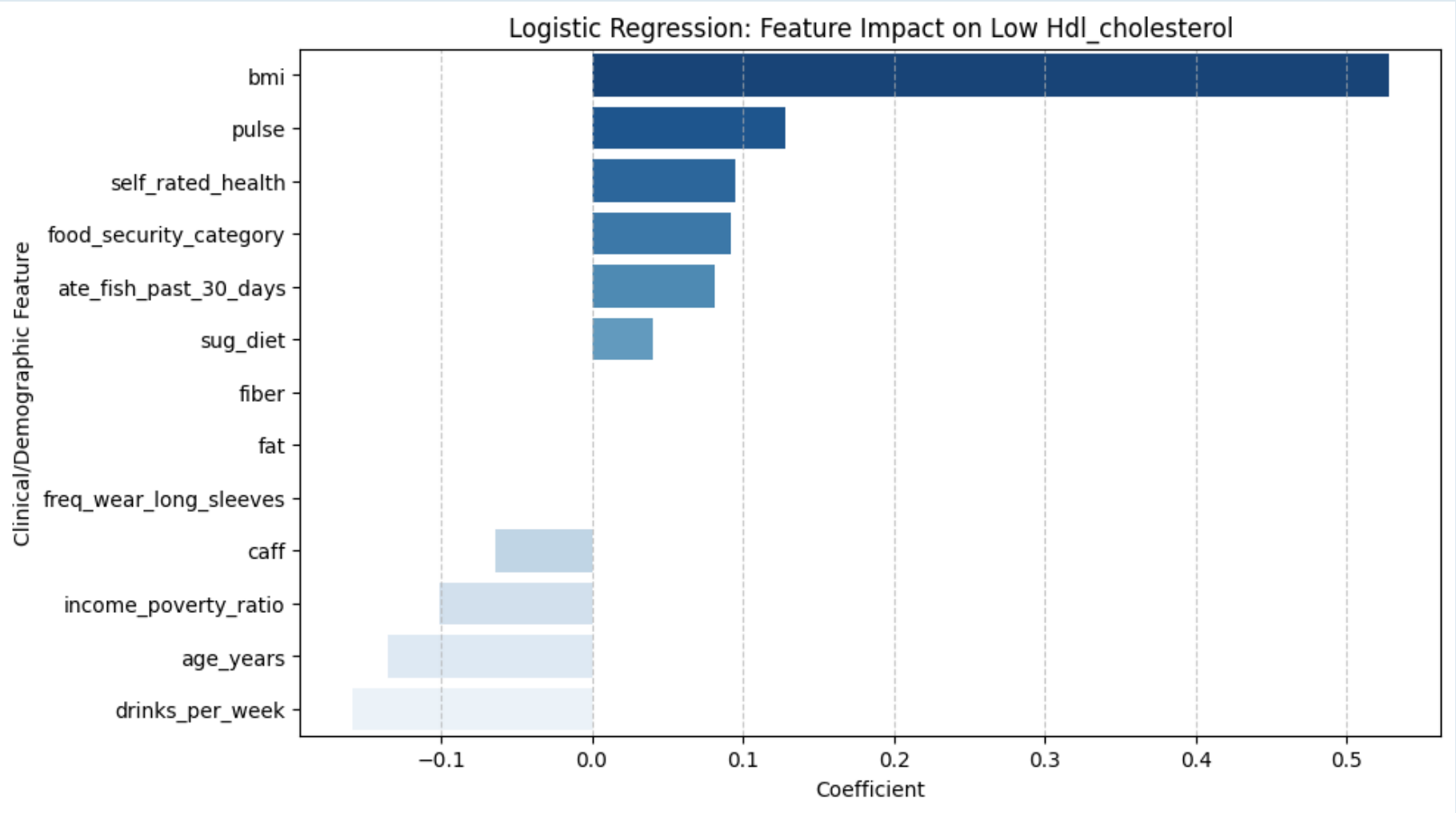


High Threshold:
Males < 40 mg/dL
Females < 50 mg/d

- Normal: 2982 (75.1%)
- Low: 987 (24.9%)
- Measures the amount of "good" cholesterol (higher levels = healthier)
- Low levels are associated with increased risk of heart attacks and strokes

<https://www.mayocliniclabs.com/test-catalog/overview/8429#clinical-and-interpretive>

Model 3: Low HDL

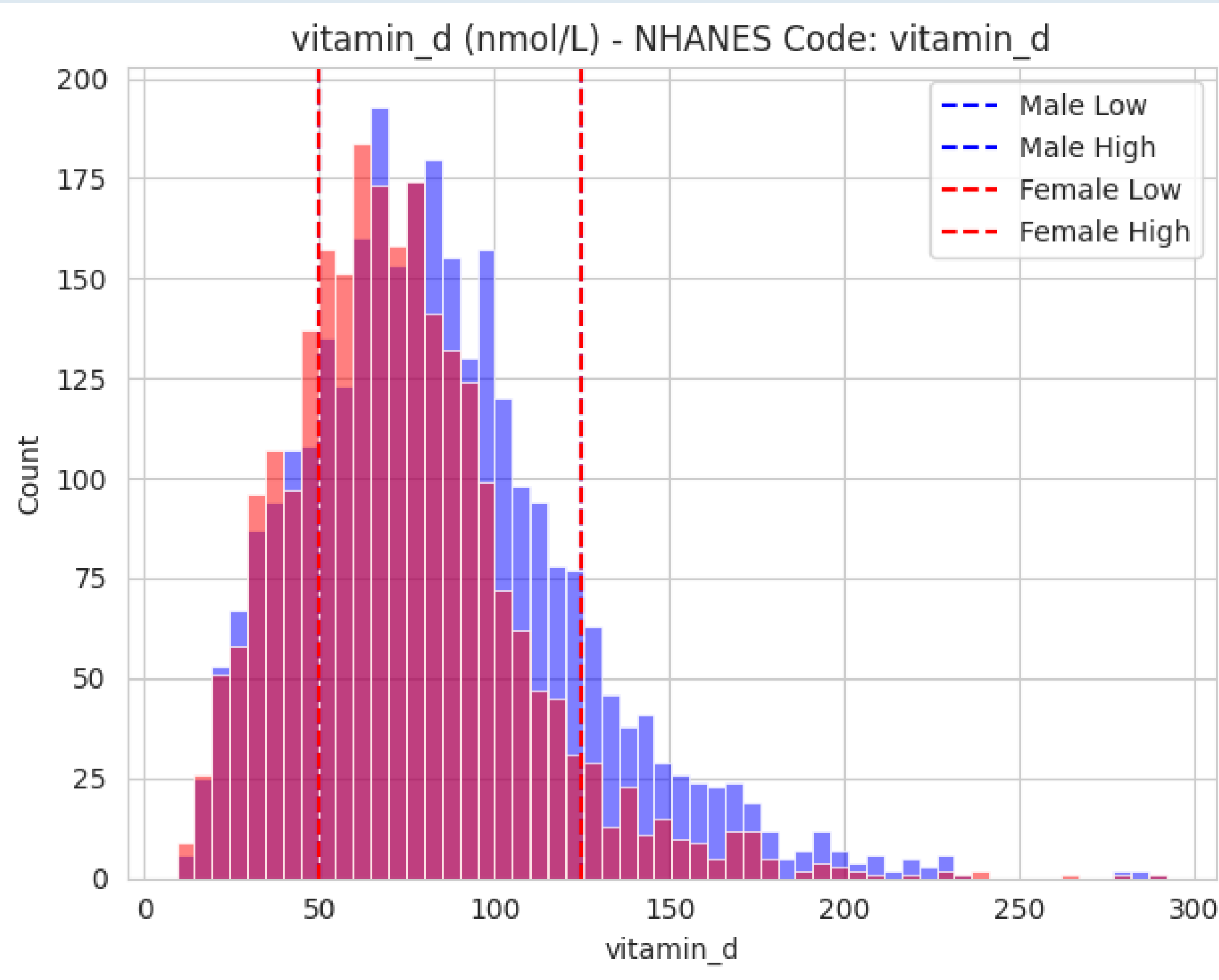


Top Predictors by odds ratio:

- BMI : 1.70
- Pulse: 1.14
- Self Rated Health: 1.10
- Food Security: 1.10
- Income to poverty ratio: 0.90

Metric	Random Forest	Logistic Regression
Discrimination & Stability		
Test AUC	0.751	0.741
Train AUC	0.821	0.742
Train-Test gap	+0.070	0.001
Detection of "Low" (positive class)		
Recall	0.72	0.64
Precision	0.43	0.42
Detection of "Normal" (negative class)		
Recall	0.69	0.71
Precision	0.88	0.86

Model 4: Low Vitamin D

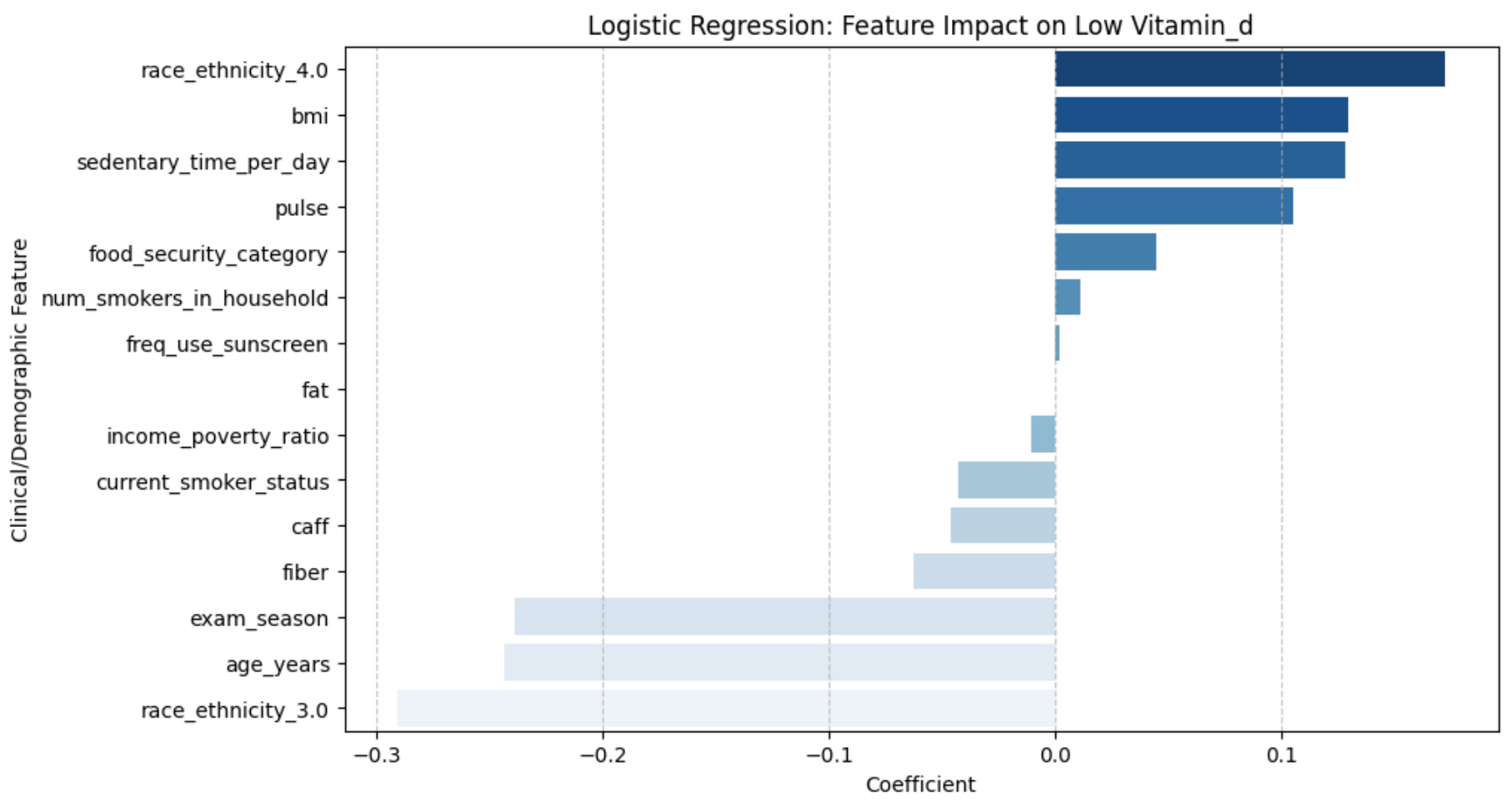


Low Threshold: 50 nmol/L

- Normal (>50): 2240 (68.1%)
- Low: 1047 (24.9%)
- Vitamin D helps absorb calcium and phosphorus for building and maintaining strong bones
- Low vitamin D is associated with brittle bones, muscle weakness, impaired immune function and increased risk of depression

<https://www.mayocliniclabs.com/test-catalog/overview/83670#clinical-and-interpretive>

Model 4: Low Vitamin D

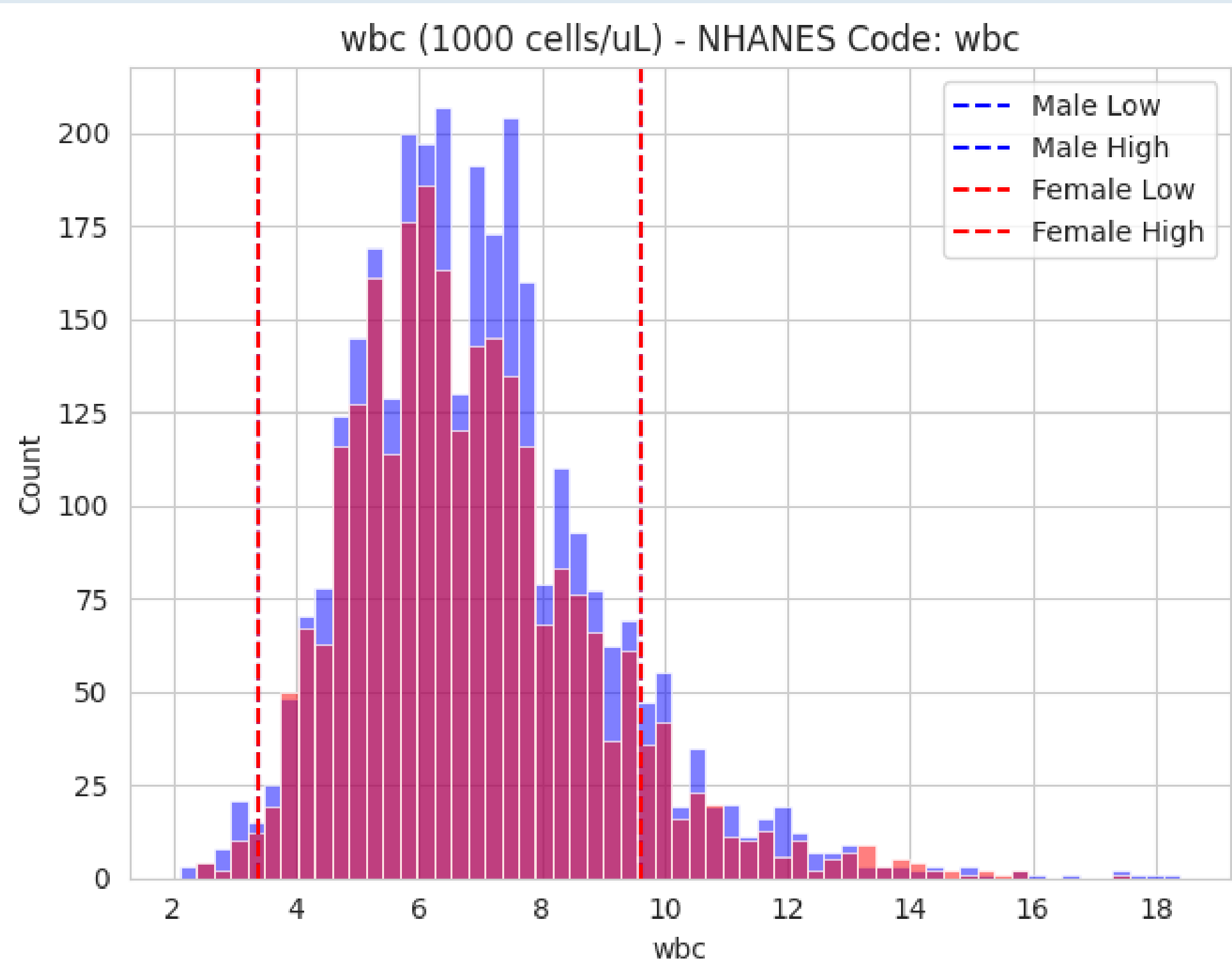


Top Predictors by odds ratio:

- Exam Season (Winter=0): 0.75
- Non –Hispanic Black: 1.19
- BMI: 1.14
- Sedentary Time: 1.14
- Pulse: 1.11

Metric	Random Forest	Logistic Regression
Discrimination & Stability		
Test AUC	0.725	0.732
Train AUC	0.813	0.738
Train–Test gap	+0.088	0.005
Detection of “Low” (positive class)		
Recall	0.62	0.64
Precision	0.51	0.53
Detection of “Normal” (negative class)		
Recall	0.72	0.73
Precision	0.80	0.81

Model 5: High WBC

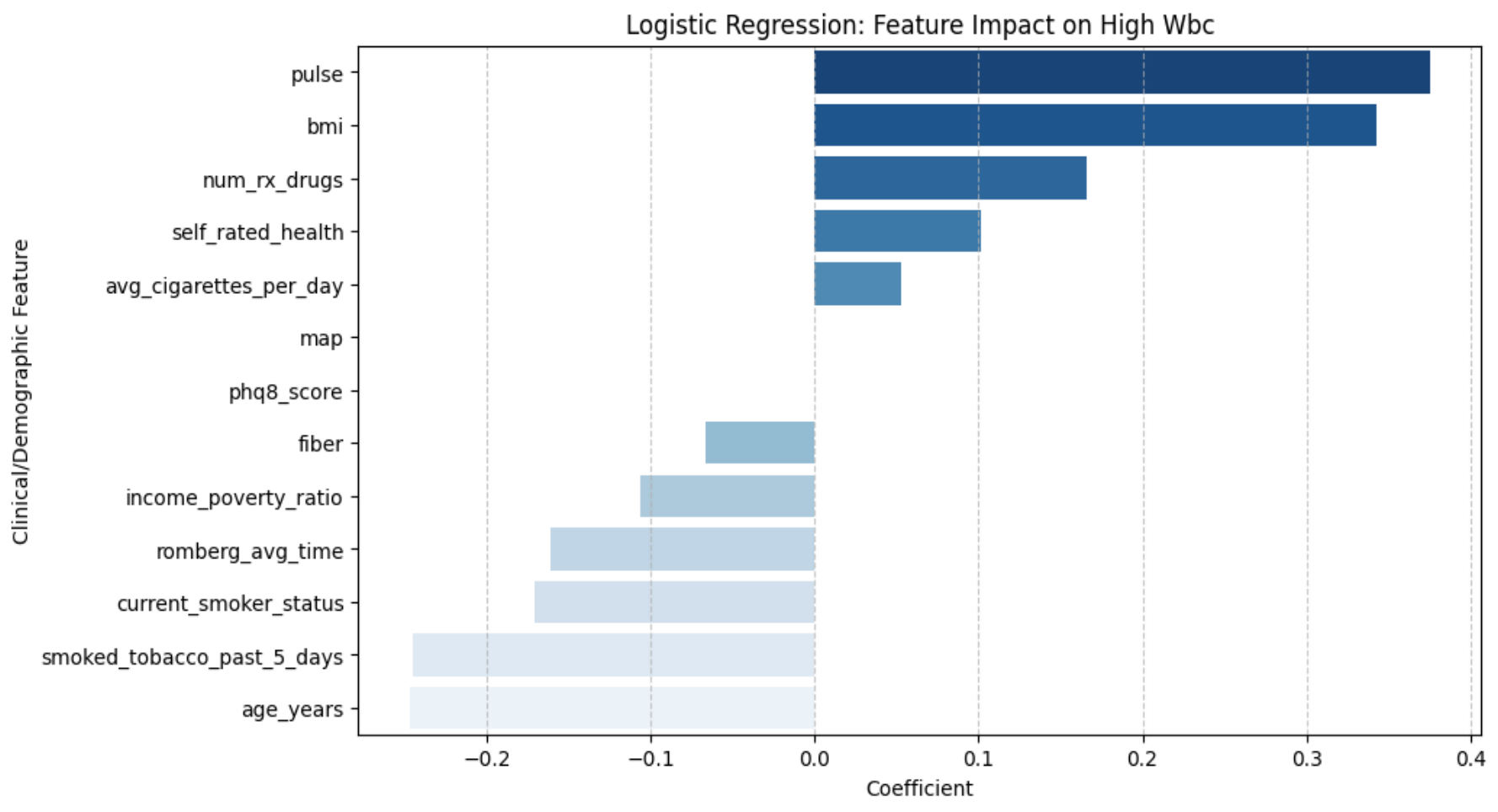


High Threshold 9.6 (1000 cells/uL):

- Normal: 5134 (90.9%)
- Low: 513 (9.1%)
- White blood cells (WBCs) help protect the body against infections, bacteria and viruses
- High levels indicate higher activities of the immune system which could be due to infections, cancers or tissue injury

<https://www.mayocliniclabs.com/test-catalog/overview/9109#clinical-and-interpretive>

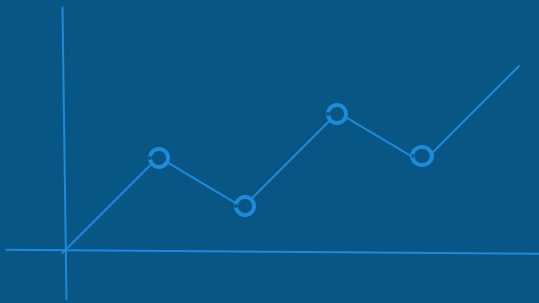
Model 5: High WBC



Top Predictors by odds ratio:

- Pulse: 1.46
- BMI: 1.41
- Number of prescription drugs: 1.18
- Age: 0.78
- Tobacco Smoking (Yes=0): 0.78

Metric	Random Forest	Logistic Regression
Discrimination & Stability		
Test AUC	0.746	0.741
Train AUC	0.827	0.748
Train-Test gap	+0.081	0.007
Detection of “High” (positive class)		
Recall	0.67	0.65
Precision	0.20	0.18
Detection of “Normal” (negative class)		
Recall	0.73	0.70
Precision	0.96	0.95



05

Recommendations and Prescriptive Insights



The background features a dark blue gradient with various geometric shapes. On the left, there are several light blue hexagons of different sizes, some with small cyan dots at their vertices. A solid dark blue circle is also present. On the right, there is a cluster of light blue hexagons, some overlapping, with a few cyan dots. Below this cluster is a small bar chart with three horizontal bars of varying lengths, the middle one being hatched. A solid dark blue circle is also located near the bottom right.

The Survey

- The top 5 models were used to create a new survey that anyone can take
- The models take these answers as inputs and output the person's risk factor for being abnormal for each test.
- [Click to take survey](#)



This tool is for informational and educational purposes only. It is not a medical diagnosis. A high-risk result means you may benefit from discussing the blood test with your doctor, not that you have a disease. Always consult a qualified healthcare provider for medical advice.

The mobile app interface shows the title "hsCRP Inflammation Risk Screener" and a subtitle "10-question survey based on NHANES population data". A yellow box contains an important notice: "Important: This tool is for informational and educational purposes only. It is not a medical diagnosis. A high-risk result means you may benefit from discussing the hsCRP blood test with your doctor, not that you have elevated inflammation. Always consult a qualified healthcare provider for medical advice." Below this is a "Health Survey" section with the instruction "Answer all 10 questions as accurately as you can. All fields are required." The first question is "1. How old are you?" with a slider ranging from 18 to 80, currently set at 50 years old. The second question is "2. How would you describe your". The app is displayed on a smartphone screen with a status bar at the top showing 1:01 and a browser address bar at the bottom showing "dpsciarr.pythonanywhere.com".

The web interface is titled "Preventive Bloodwork Screening Tool" and is "Powered by NHANES population data". A yellow box contains a privacy notice: "This tool does not collect, store, or transmit any personal information. No data you enter is saved, shared, or sold. All calculations are performed locally and discarded when you leave the page. This tool is for informational purposes only and is not a medical diagnosis." Below this is a "SELECT TESTS" section with a "Select all" link and five test options, each with an unchecked checkbox: "hsCRP" (A marker of systemic inflammation), "HbA1c" (A measure of average blood sugar over 3 months), "Vitamin D" (Supports bone health and immune function), "White Blood Cell Count" (An indicator of immune system activity), and "HDL Cholesterol" (The protective form of cholesterol). A blue "Take Survey →" button is at the bottom.

Limitations



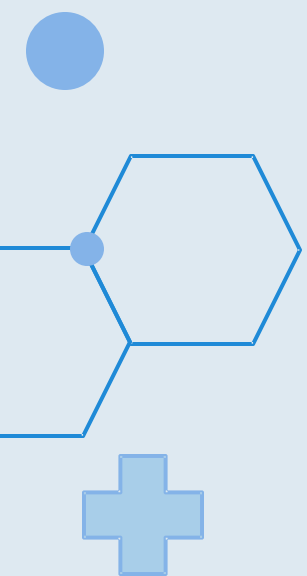
Survey Data

- The majority of the data is based on questionnaires that rely on the respondents' recall which may not always be accurate.
- Social Desirability Bias



Laboratory Data

- Many other factors, like genetics, can determine a person's blood composition outside of those tested in this project
- False positive and negative blood results can occur randomly or be caused by improper specimen handling and preparation

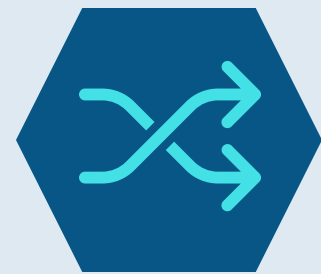


Ways to Improve



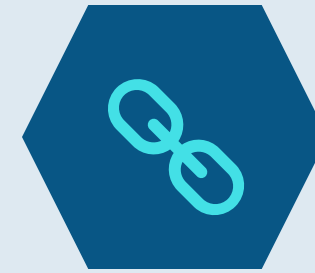
Increase sample size

Adding data from other cycles would help reduce variance and increase generalizability



Resampling Methods

Methods like SMOTE would help address the class imbalance



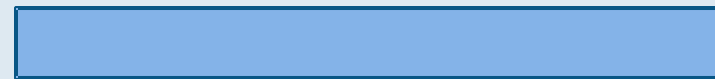
Ensemble Methods

Use a stacking or boosting method to compensate errors made in random forest and logistic regression

Future Direction

Release the survey to the general public to increase health awareness

Survey Testing



01

Set up a trial to test survey predictions against real clinical laboratory tests for select respondents

Clinical Trials



02

Implement similar surveys in primary care offices to streamline new patient visits

Healthcare Deployment



03

Ethical Considerations



Socioeconomic and Demographic Questions

Certain demographic and socioeconomic factors, like race and income/poverty, are used only for their clinical relevance to abnormal tests



Protection of Health Information

All user information collected on the app is used strictly for screening and educational purposes. No data is stored and surveys are completely anonymous.

Resources

Acta Scientific. (n.d.). *[Title of article]*. *Acta Scientific Medical Sciences*.

<https://actascientific.com/ASMI/pdf/ASMI-05-1153.pdf>

Best Practice Advocacy Centre New Zealand (BPACNZ). (2015, April). *Laboratory investigations: A guide for primary care*. <https://bpac.org.nz/BT/2015/April/docs/BT26-laboratory-investigations.pdf>

Elsevier. (n.d.). *Social desirability bias*. ScienceDirect Topics.

<https://www.sciencedirect.com/topics/social-sciences/social-desirability>

Mayo Clinic Laboratories. (n.d.). *Mayo Clinic Laboratories*. <https://www.mayocliniclabs.com/>



Thank You

Questions?

